Anshuman Kumar

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Skills

Programming Languages	: C, C++11/14/17, Rust
Scripting Languages	: BASH, Python
Frameworks/Libraries	: POSIX, GTK, GStreamer, XDEVapi, Slint
Other Libraries	: Arduino, MAVLink
Build System & Tools	: Make, GDB, LLDB
Other Tools	: VS Code, Dia

Work Experience

Internship at IDR Research And Development Pvt Ltd.

1 Wi-Fi based Mesh Network

- Built a dynamic **multiple role-based multi-node Network** implemented over ESP32
- Developed custom protocol with packet drop detection, checksum validation
- Added **healing feature** to combat master node failures during operation

2 LoRa Configurator

- Built an UART-based easy-to-use LoRa configuration library
- Used modern C++ practices & design techniques to improve maintainability

3 Flight Controller communication Interface

- Reduced the packet size and transmission time over large distance
- Used **MAVLink protocol** over ESP32 board as intermediary transmitter for FC to ease implementing partial Drone Swarm
- Implemented **User Action mapping interface** to map custom User-defined OTA commands into targeted Drone actions

My Projects

1 SysInfo — TheNoobCoder-8513/SysInfo

- A **Modern System Resource Monitor** for daily-users
- Powered By Rust's memory-safety guarantees and Slint for GUI
- GUI is built from scratch in Slint's DSL, writing own each higher level component's design and functionality, such as UsageChart, InfoCards, and more.

2 ConnecTupp

- Chat application for Linux written entirely in C from scratch
- Built a socket-based chat system with user/group messaging
- Designed a lightweight custom protocol and **GTK-based UI**
- Learnt implementation of data transmission techniques, syscalls and multithreading

3 Connect — TheNoobCoder-8513/Connect

- Fast, minimalist **Phonebook** written in C
- Plain text file based storage for file operations
- Learnt working of low-level filesystem, input events and Data Structures for efficient storage

4 VirtAudio Cable

- **Virtual Audio router** for linux written using C++ With UI built using **Qt**
- Learnt about pipes, streams and Linux audio management
- Enabled Users to merge multiple audio inputs into one virtual input source and use it as input source in single audio input supported applications

Education

B.Tech (CSE) – ITS Engineering College	2023 – Present
Intermediate – K.D.S. College	2021 - 2023
High School – DAV Public School	2021